

XINYU MAO

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RESEARCH INTEREST

My research focuses on the foundations of cryptography. More recently, I have also become interested in real-world cryptography and AI safety.

EDUCATION

University of Southern California

Aug. 2022 – Aug. 2026

Ph.D. in Computer Science

Advisor: [Prof. Jiapeng Zhang](#)

Los Angeles, United States

Shanghai Jiao Tong University

Aug. 2018 – June 2022

Bachelor of Engineering in Computer Science

A member of ACM Honor Class, Zhiyuan College

Advisors: [Prof. Yijia Chen](#) and [Prof. Yong Yu](#)

GPA: 3.96 / 4.3 | Ranking: 2 / 33

Shanghai, China

RESEARCH EXPERIENCE

University of Southern California

May 2021 - April 2022

Research Intern advised by [Prof. Jiapeng Zhang](#)

Research Topic: foundations of cryptography.

Shanghai Jiao Tong University

Sept. 2020 – Feb. 2021

Undergraduate Researcher advised By [Prof. Shengli Liu](#)

Research Topic: public-key cryptography

PUBLICATIONS

1. Xinyu Mao, Noam Mazon, and Jiapeng Zhang. Non-adaptive universal one-way hash functions from arbitrary one-way functions. In *Advances in Cryptology – EUROCRYPT 2023: 42nd Annual International Conference on the Theory and Applications of Cryptographic Techniques, Lyon, France, April 23-27, 2023, Proceedings, Part IV*, page 502–531, Berlin, Heidelberg, 2023. Springer-Verlag
2. Mi-Ying Huang, Xinyu Mao, Guangxu Yang, and Jiapeng Zhang. Communication lower bounds of key-agreement protocols via density increment arguments. In Guy Rothblum and Hoeteck Wee, editors, *Theory of Cryptography*, pages 185–206, Cham, 2023. Springer Nature Switzerland
3. Xinyu Mao and Jiapeng Zhang. On the power of svd in the stochastic block model. *Advances in Neural Information Processing Systems*, 36:32765–32781, 2023
4. Xinyu Mao, Guangxu Yang, and Jiapeng Zhang. Gadgetless lifting beats round elimination: Improved lower bounds for pointer chasing. *Innovations in Theoretical Computer Science (ITCS)*, 2025
5. Yilei Chen and Xinyu Mao. Universal computational extractors and multi-bit aipo from lattice assumptions. In Serge Fehr and Pierre-Alain Fouque, editors, *Advances in Cryptology – EUROCRYPT 2025*, pages 63–93, Cham, 2025. Springer Nature Switzerland
6. Mi-Ying Miryam Huang, Xinyu Mao, Shuo Wang, Guangxu Yang, and Jiapeng Zhang. A min-entropy approach to multi-party communication lower bounds. In *40th Computational Complexity Conference (CCC 2025)*, pages 33–1. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2025

7. Yu Chen, Xinyu Mao, and Hongxu Yi. Advancing pseudorandom codes: Beyond parity checks and standard-model CCA1 security. Cryptology ePrint Archive, Paper 2026/1228, To appear in Asiacrypt, 2026
8. Mi-Ying Miryam Huang, Xinyu Mao, and Jiapeng Zhang. Sublinear proofs over polynomial rings. Cryptology ePrint Archive, Paper 2025/199, To appear in Asiacrypt, 2025
9. Xinyu Mao. Finding random collisions for random degree-2 functions. Cryptology ePrint Archive, Paper 2026/1044, To appear in TCC, 2026

HONORS AND AWARDS

Zhiyuan Outstanding Student Scholarship	2022
Shanghai Outstanding Graduate	2022
Foresight-Sequoia Scholarship 70 winners in SJTU each year	2021
Huawei Scholarship 14 winners at Zhiyuan College each year	2020
Zhiyuan Honorary Scholarship Top 5% in SJTU each year	2018, 2019, 2020
Bronze Medal The 34th China National Olympiad in Informatics (NOI)	2017

TEACHING EXPERIENCE

CSCI 575: Quantum Computing and Quantum Cryptography Teaching Assistant	Fall 2025 University of Southern California
CSCI 531: Applied Cryptography Teaching Assistant	Spring 2025 University of Southern California
CSCI 270: Introduction to the Theory of Computation Teaching Assistant	Fall 2023 University of Southern California
Great Ideas in Computer Science Teaching Assistant I wrote an <u>Enigma Simulator</u> to facilitate teaching.	Fall 2019, Fall 2020, Fall 2021 Shanghai Jiao Tong University

SKILLS

Programming: C++, Java, Python, JavaScript, Latex
Language: Chinese (native), English (TOEFL 105)